Course Name: Ethical Hacking

Assignment - Week 1

Which of the following belong to the CIA Triad in information security?

- **a.** Confidentiality
- **b.** Authentication
- **c.** Availability
- **d.** Accountability
- **e.** Integrity
- **f.** Authorization

Correct Answer: a, c, e

Which of the following penetration testing models provides the tester with no prior information about the target network?

- **a.** White box model
- **b.** Black box model
- **c.** Gray box model
- **d.** Blue box model

Correct Answer: b

Which of the following statement(s) is/are true for packet-switched networks?

- **a.** Bandwidth is dedicated for the entire duration of a session.
- **b.** Multiple connections can share the same communication link.
- **c.** Delay is always constant regardless of network load.
- **d.** It requires end-to-end circuit setup before data transfer.
- **e.** Data is divided into smaller packets before transmission.
- **f.** It guarantees zero packet loss under all conditions.

Correct Answer: b, e

Which OSI layer is primarily responsible for framing, MAC addressing, and error detection over a physical link?

- **a.** Physical layer
- **b.** Data link layer
- **c.** Network layer
- **d.** Transport layer
- **e.** Session layer

Correct Answer: b

A 1500-byte packet is sent over a 100 Kbps point-to-point link whose propagation delay is 8 msec. The packet will reach the destination after _____ msec.

Correct Answer: 120 to 130 msec (Assume 1K=1000)

Detailed Solution:

- Transmission rate = 100,000 bits per second.
- 1500 bytes = 12,000 bits.
- Transmission time = 12,000 / 100,000 = 0.12 sec = 120 msec.
- Total time = 120 + 8 = 128 msec.

What is the primary purpose of an IP address in a network?

- **a.** It identifies the physical hardware manufacturer of a device.
- **b.** It uniquely identifies a host or interface on a network.
- **c.** It specifies the number of open ports on a system.
- **d.** It determines the encryption algorithm used by a host.

Correct Answer: b

_____ is the minimum size (in bytes) of a TCP header without options.

Correct Answer: 20

Detailed Solution: The minimum size of a TCP header is 20 bytes when no options are used. Additional options can increase this size up to 60 bytes. Thus, the correct answer is 20.-----

Which IP header field is decremented at every router hop and is primarily used to prevent packets from looping indefinitely in a network?

- **a.** Header checksum
- **b.** Fragment offset
- **c.** Time to Live (TTL)
- **d.** HLEN
- **e.** Identification

Correct Answer: c

If the IP header is 224 bits long, what will be the value (in decimal) of the HLEN field?

Correct Answer: 7

Detailed Solution: HLEN represents the header length in multiples of 32 bits (4 bytes). Thus, 224 bits = 7 × 32 bits. Hence HLEN = 7. Thus the correct answer is 7.-----

_____ bits are used to represent a MAC address.

Correct Answer: 48

Detailed Solution: A MAC address (data link layer / hardware address) is 48 bits long, typically represented as 12 hexadecimal digits. Thus, the correct answer is 48.

Assignment - 2

An IP packet arrives with the first eight bits as 0100 0101. How many bytes are there in the OPTIONS field?

- **Correct Answer:** 0
- **Detailed Solution:** First four bits (0100) = version 4. Next four bits (0101 = 5) = header length. Header length = $5 \times 4 = 20$ bytes, which is the minimum IP header size with no options. Thus, the OPTIONS field size = $20 - 20 = 0$ bytes.

Which of the following is/are true for TCP header flag bits?

- a. SYN=1, ACK=0 indicates a connection request.
- b. SYN=1, ACK=1 indicates connection confirmation.
- c. FIN flag is used to reject a connection.
- d. RST flag is used to gracefully terminate a connection.
- e. URG flag indicates urgent data in the segment.
- **Correct Answer:** a, b, e
- **Detailed Solution:** SYN=1, ACK=0 is a connection request; SYN=1, ACK=1 is connection confirmation; URG flag indicates urgent data. FIN is used for graceful connection termination (not rejection), and RST is used to abruptly reset/reject a connection (not graceful termination).

Which of the following represents a valid broadcast address for its respective subnet (as specified)?

- a. 10.255.255.255 for network 10.0.0.0/8
- b. 10.20.0.0 for network 10.20.0.0/16

- c. 192.168.5.255 for network 192.168.5.0/24
- d. 172.16.255.0 for network 172.16.0.0/16
- e. 192.168.5.0 for network 192.168.5.0/24
- **Correct Answer:** a, c

An IP packet arrives at the final destination with D flag = 0, M flag = 0, and offset = 5. Which of the following is true?

- a. The packet has not been fragmented.
- b. The packet is the first fragment.
- c. The packet is the last fragment of a fragmented packet.
- d. The packet will be dropped due to invalid flags
- **Correct Answer:** c

Which of the following IP address ranges are reserved for private use?

- a. 10.0.0.0 to 10.255.255.255
- b. 100.64.0.0 to 100.127.255.255
- c. 172.16.0.0 to 172.31.255.255
- d. 192.168.0.0 to 192.168.255.255
- e. 169.254.0.0 to 169.254.255.255
- f. 224.0.0.0 to 239.255.255.255
- **Correct Answer:** a, c, d

Which of the following well-known port number(s) is/are correctly matched with their corresponding service?

- a. HTTP - 80
- b. FTP - 21
- c. SSH - 25
- d. DNS - 23
- e. Telnet - 53
- f. SMTP - 22
- **Correct Answer:** a, b

What is the subnet mask corresponding to CIDR notation /26?

- a. 255.255.255.0
- b. 255.255.255.128
- c. 255.255.255.192
- d. 255.255.255.224
- e. 255.255.255.240
- **Correct Answer:** c

Which of the following statement(s) is/are true about IPv4 fragmentation and the "Don't Fragment" (DF) bit?

- a. If DF=1 and packet size exceeds MTU, the packet is dropped and an error is generated.
- b. If DF=1, the router fragments the packet regardless of size.
- c. If DF=0, fragmentation is allowed if necessary.
- d. DF bit is part of the Fragment Offset field.
- e. DF=1 guarantees the packet will always reach its destination.
- **Correct Answer:** a, c

An IP packet has HLEN = 12 and TOTAL LENGTH = 900 bytes. The number of data bytes in the packet will be _____

- **Correct Answer:** 852
- **Detailed Solution:** HLEN = 12 means header size = $12 \times 4 = 48$ bytes. Data bytes = $900 - 48 = 852$ bytes.

_____ is the size (in bytes) of a UDP header.

- **Correct Answer:** 8
- **Detailed Solution:** The UDP header has 4 fields (source port, destination port, length, checksum), each of 2 bytes, totaling 8 bytes.

Assignment - 3

Which of the following statement(s) is/are true regarding connectionless routing used by the IP protocol?

- **a.** Each packet is treated as an independent unit for routing decisions.
- **b.** A fixed path is established before any packet is transmitted.
- **c.** Different packets between the same source and destination may follow different paths.
- **d.** IP protocol uses connection-oriented routing.

Correct Answer: a, c

Which of the following statement(s) is/are false regarding direct and indirect packet delivery?

- **a.** Direct delivery occurs when the destination host and deliverer are on the same physical network.
- **b.** Indirect delivery occurs when the destination host is not on the same network as the deliverer.
- **c.** In direct delivery, packets always pass through at least one router.
- **d.** In indirect delivery, the packet is routed hop by hop until it reaches a router on the destination's network.
- **e.** Direct delivery can occur between the last router and the destination host.

Correct Answer: c

Which routing table flag indicates that a route is specific to a single host rather than an entire network?

- **a.** U
- **b.** G
- **c.** H
- **d.** D
- **e.** M

Correct Answer: c

In the OSPF routing protocol, which type of packet is used to determine whether a neighboring router is active?

- **a.** Link State Update
- **b.** Link State Request
- **c.** Hello packet
- **d.** Link State Acknowledgement
- **e.** TCP SYN packet

Correct Answer: c

Which type of address should be used when a packet needs to be delivered to every host within a specific network?

- **a.** Unicast address

- **b.** Broadcast address
- **c.** Anycast address
- **d.** None of these

Correct Answer: b

Which of the following statement(s) is/are false regarding routing protocols?

- **a.** RIP uses distance vector routing and can suffer from the count-to-infinity problem.
- **b.** OSPF is a link-state routing protocol.
- **c.** BGP is used for routing within a single autonomous system.
- **d.** OSPF uses Dijkstra's algorithm to compute shortest paths.
- **e.** RIP is typically used for routing between different autonomous systems.

Correct Answer: c, e

What is the address used to represent the default route in a routing table (in IPv4)?

- **a.** 127.0.0.1
- **b.** 255.255.255.255
- **c.** 0.0.0.0
- **d.** 192.0.0.1
- **e.** None of these

Correct Answer: c

Which of the following statement(s) is/are false regarding IPv6?

- **a.** IPv6 supports address classes similar to IPv4 (Class A, B, C).
- **b.** IPv6 addresses are typically represented in hexadecimal, separated by colons.
- **c.** IPv6 eliminates the need for fragmentation by intermediate routers.
- **d.** The base header size of IPv6 is variable, ranging from 20 to 60 bytes.

Correct Answer: a, d

In IPv6 base header, how many bits are used for Hop Limit?

Correct Answer: 8

An organization is assigned 172.20.64.0/20 network block. What number of valid host addresses are available in this block?

Correct Answer: 4094

Detailed Solution: A /20 prefix leaves 12 host bits ($32 - 20 = 12$), giving $2^{12} = 4096$ addresses. Subtracting the network address and broadcast address leaves 4094 usable host addresses.

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Assignment - 3

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ASSIGNMENT 5

Which of the following NMAP option(s) can trigger execution of NMAP scripts?

a. -A
b. -sC
c. -Pn
d. --script=default
e. -sn

Correct Answer: a, b, d

Which of the following tool(s)/software CANNOT be used for vulnerability scanning?

a. Nessus
b. Hydra
c. Nexpose
d. Crunch
e. NMAP

Correct Answer: b, d

Which of the following is NOT considered a common password cracking approach?

a. Dictionary attack
b. Brute-force attack
c. Rule-based attack
d. Packet sniffing
e. Shoulder surfing

Correct Answer: d

Which protocol in TCP/IP is connectionless and does not provide reliability?

a. TCP
b. UDP
c. FTP
d. TELNET

Correct Answer: b

Which tool is primarily used to generate custom wordlists for use in dictionary-based password attacks?

a. Hydra

b. Nessus
c. Crunch
d. Wireshark
e. Hashcat
Correct Answer: c
Which of the following statement(s) is/are false regarding malware?
a. All malware types are capable of self-replication.
b. Malware can grant partial or full control of a system to an attacker.
c. Malware can be delivered through file sharing or fake/pirated software.
d. Malware can modify, corrupt, or delete data on infected systems.
e. None of these.
Correct Answer: a
In a MAC flooding attack, what is the primary goal of overwhelming a switch's CAM table?
a. To crash the switch permanently.
b. To force the switch to behave like a hub, broadcasting traffic to all ports.
c. To increase the switch's available memory.
d. To disable the switch's power supply.
e. None of these.
Correct Answer: b
Which of the following NMAP script(s) is/are used specifically for identifying the operating system of an SMB-enabled target?
a. smb-os-discovery
b. smb-brute
c. http-os-brute
d. smb-enum-users
e. ssh-brute.
Correct Answer: a
An attacker is using hydra to perform a dictionary attack on an SSH service, testing a wordlist of 5000 passwords against 4 different usernames. The maximum number of logins attempts that could be made is _____.
Correct Answer: 20000
Detailed Solution: Total attempts = number of usernames x number of passwords = 4 x 5000 = 20,000 (assuming every password is tried against every username, worst case before a match is found).
Thus, the correct answer is 20000.
If a symmetric-style brute-force password cracker needs to try all combinations of a 4-character password using only lowercase letters (26 possible characters per position), how many total password combinations are possible?
Correct Answer: 456976
Detail Solution: Total combinations = 26^4 = 26 x 26 x 26 x 26 = 456,976.
Thus, the correct answer is 456976.

Course Name: Ethical Hacking

Assignment- Week 6

Which of the following attack types involves an attacker pretending to be a legitimate entity to gain unauthorized access?

a. Replay attack
b. Masquerade attack
c. Modification attack
d. Denial of service attack
e. Traffic analysis

Correct Answer: b

Encrypt the plaintext "SECURE" using a substitution cipher where each letter is replaced by its k-th next letter, assuming alphabets wrap around (Z followed by A) and k = 3.

a. VHFXUH
b. VFEXRU
c. UHFXVH
d. VHFXRV
e. None of these.

Correct Answer: a

Consider a mono-alphabetic cipher with the key mapping: (A B C D E F G H I J K L M N O P Q R S T U V W X Y Z) mapped to (Q W E R T Y U I O P A S D F G H J K L Z X C V B N M). What will be the encrypted form of the message "CODE"?

a. EGRT
b. EGDT
c. EODT
d. EGDR
e. None of these

Correct Answer: a

Detailed Solution: Using the given mapping: C→E, O→G, D→R, E→T. Hence, "CODE" encrypts to "EGRT".

Which of the following statement(s) is/are false regarding hashing algorithms used for message integrity?

a. A good hash function should be resistant to collisions.
b. The same input will always produce the same hash output.
c. A hash function can be reversed to retrieve the original message.
d. Even a small change in input should produce a significantly different hash output.

Correct Answer: c

Detailed Solution: A good cryptographic hash function is collision-resistant, deterministic (the same input gives the same output), and exhibits the avalanche effect (small input changes drastically alter the output). Hash functions are one-way functions and cannot be reversed to retrieve the original input.

Decrypt the ciphertext "KHOOR" which was encrypted using a substitution cipher with a shift of k = 3 (each letter replaced by its k-th next letter during encryption, the alphabets wrap around).

a. HELLO
b. WORLD
c. HELPS

d. HFLLO
e. None of these
Correct Answer: a
Detailed Solution: To decrypt, shift each letter back by 3: K→H, H→E, O→L, O→L, R→O.
This gives "HELLO".
Thus, the correct option is (a).

A DoS attack that makes a system or service inaccessible to legitimate users falls under which category of security attack?

a. Interruption
b. Interception
c. Modification
d. Fabrication
e. Replay
Correct Answer: a

For encryption using public-key cryptography, if sender A wants to send a confidential message to receiver B, which key is used to encrypt the message?

a. A's private key
b. A's public key
c. B's public key
d. B's private key
Correct Answer: c

_____ is the block size (in bits) used for plaintext blocks in the Data Encryption Standard (DES) algorithm.

Correct Answer: 64
Detailed Solution: DES processes plaintext in fixed 64-bit blocks, using a 56-bit key (with 8 additional parity bits, making the stored key 64 bits, though only 56 bits are effectively used for encryption).

150 parties want to exchange messages securely using a public key encryption technique like RSA. How many distinct key values will be required?

Correct Answer: 300
Detailed Solution: In asymmetric (public-key) encryption, each party holds two keys (public and private). For N parties, total keys = 2N. For N = 150, 2 x 150 = 300.

40 parties want to exchange messages securely using a symmetric key encryption algorithm. How many distinct keys will be required?

Correct Answer: 780
Detailed Solution: In symmetric encryption, every pair of communicating parties needs a separate key. For N parties, the number of keys = NC2 = N(N-1)/2. For N = 40, 40 x 39 / 2 = 780.

Assignment- Week 7

A message M is fed to a hash function to generate hash value H = HASH(M). Which of the following statement(s) is/are true?

a. The number of bits in H is typically much smaller than the number of bits in M.
b. The number of bits in H and M are always exactly equal.
c. Hashing produces a fixed-length output regardless of input size.

d. The output length of a hash function grows proportionally with the input length.

Correct Answer: a, c

Which of the following statement(s) is/are true?

a. The block size of DES is 64 bits.

b. The key size of DES is 128 bits.

c. The hash digest size of MD5 is 128 bits.

d. SHA-256 produces a 512-bit digest.

e. Triple DES uses a fixed 512-bit key.

Correct Answer: a, c

Which of the following statement(s) is/are true regarding SSL/TLS protocols?

a. HTTPS relies on SSL/TLS to secure HTTP communication.

b. SSL/TLS guarantees faster data transmission compared to plain HTTP.

c. TLS operates at the physical layer of the network stack.

d. The SSL Record Protocol guarantees non-repudiation of the sender.

Correct Answer: a

Which of the following statement(s) is/are true regarding collision resistance in hash functions?

a. It is a weaker requirement than second preimage resistance.

b. It should be computationally difficult to find any two distinct messages producing the same hash.

c. Collision resistance guarantees encryption confidentiality.

d. Collision resistance is trivial to achieve for any hash function design.

e. Collision resistance applies only to keyed hash functions (MACs).

Correct Answer: b

Which of the following statement(s) is/are true regarding digital signature verification?

a. The verifier uses the signer's public key to verify a signature.

b. The verifier uses the signer's private key to verify a signature.

c. Successful verification confirms the message has not been altered and was signed by the claimed signer.

d. Verification requires the signer to be online and actively participating.

Correct Answer: a, c

Which property of digital signatures ensures that the original signer cannot later deny having signed a message?

a. Confidentiality

b. Integrity

c. Non-repudiation

d. Availability

e. Authentication

Correct Answer: c

Which of the following statements is false regarding keyed vs unkeyed hash functions?

a. An unkeyed hash function can be used to verify data integrity when the hash value is obtained from a trusted source.

b. A keyed hash function requires a secret key as part of its computation.

c. An unkeyed hash function inherently provides authenticity.

d. HMAC can be used to provide both integrity and a degree of authentication.

Correct Answer: c

In IPSec, which mode encrypts/protects only the payload (transport layer data) while keeping the original IP header intact?

a. Tunnel mode
b. Transport mode
c. Replay mode
d. Confidential mode
e. None of these

Correct Answer: b

Hash digest size produced by the MD5 algorithm is _____ bits.

Correct Answer: 128

The block size used by the SHA-512 hashing algorithm is _____ bits.

Correct Answer: 1024

Detailed Solution: SHA-512 processes the input message in blocks of 1024 bits, producing a 512-bit hash digest.
Thus, the correct answer is 1024.

Assignment- Week 8

Which of the following statements best describes steganography?

a. Encrypting information using public and private keys.
b. Embedding information within another cover medium.
c. Tunnelling data using HTTPS and TLS.
d. Compressing data to reduce its transmission size.
e. None of these.

Correct Answer: b

Which of the following are valid examples of steganography?

a. Hiding text data within an image file.
b. Hiding secret data within an audio clip.
c. Encrypting a file so only the recipient can decrypt it.
d. Hiding data within an executable file.
e. Digitally signing a document to prove authorship.
f. None of these.

Correct Answer: a, b, d

Which of the following is/are examples of behavioural biometrics?

a. Fingerprint
b. Signature dynamics
c. Gait
d. Iris pattern
e. Keystroke dynamics
f. Retina pattern
g. Facial structure

Correct Answer: b, c, e

Which biometric modality is generally considered to offer very high uniqueness and reliability for identification?

a. Voice recognition
b. Facial recognition
c. Iris scan

d. Signature dynamics
e. Gait analysis
Correct Answer: c

What is a Denial-of-Service (DoS) attack?

a. An attack that deletes or modifies files stored on a system.
b. An attack that prevents legitimate users from accessing a service.
c. An attack that specifically targets and destroys stored password databases.
d. None of these.
Correct Answer: b

How does a Smurf DoS attack typically work?

a. It sends oversized IP packets to crash the target host.
b. It sends forged ICMP Echo requests to a network's broadcast address with the victim's spoofed source IP, causing many hosts to flood the victim with replies.
c. It exploits buffer overflow vulnerability in web server software.
d. It uses SQL injections to compromise a backend database.
e. None of these.
Correct Answer: b

Which of the following attacks exploits the TCP three-way handshake by leaving connections in a half-open state?

a. Ping of Death attack
b. SYN flooding attack
c. Smurf attack
d. DNS spoofing attack
Correct Answer: b

What does PGP (Pretty Good Privacy) primarily provide for secure email communication?

a. Authentication of the sender
b. Confidentiality of the message content
c. Guaranteed message availability
d. Protection against email spam filtering
Correct Answer: a, b

What is the primary purpose of the Domain Name System (DNS)?

a. To provide end-to-end encryption for email communication.
b. To translate human-readable domain names into IP addresses.
c. To compress and fragment large network messages.
d. To hide secret data inside cover media files.
e. None of these.
Correct Answer: b

Consider a grayscale image of size 1500 x 1200, where each pixel is stored using 8 bits. Using LSB steganography (1 bit hidden per pixel), how many bytes of information can be hidden in this image?

Correct Answer: 225000

Detailed Solution: Total pixels = 1500 x 1200 = 1,800,000. Since each pixel allows hiding 1-bit, total hidden bits = 1,800,000 bits. Converting to bytes: 1,800,000 / 8 = 225,000 bytes.

Course Name: Ethical Hacking

Assignment- Week 9

Which of the following statement(s) is/are true for sniffing?

a. It is a process of analyzing network activity by capturing network traffic.
b. It is a process of finding vulnerabilities in a network.
c. It is a process used for user enumeration.
d. It may require the NIC to operate in promiscuous mode to capture frames not specifically addressed to the host.
e. None of these.

Correct Answer: a, d

Which of the following is/are not a feature of Wireshark?

a. Capturing and analyzing network packets.
b. Generating statistical reports on captured traffic.
c. Manipulating and modifying live network traffic.
d. Displaying packet details with GUI support.
e. Carrying out SQL injection attacks.
f. Filtering captured packets by protocol.

Correct Answer: c, e

Which of the following protocols are vulnerable to sniffing attacks?

a. HTTP
b. FTP
c. Telnet
d. HTTPS
e. SSL
f. SSH

Correct Answer: a, b, c

Which of the following feature(s) is/are not present in the Ettercap tool?

a. IP-based and MAC-based filtering.
b. Character injection into live connections.
c. Packet filtering and dropping.
d. SQL injection automation.
e. Password cracking via dictionary attack.

Correct Answer: d, e

Which of the following is not a human-based social engineering attack?

a. Impersonation
b. Shoulder surfing
c. Piggybacking/tailgating
d. Phishing email

Correct Answer: d

A DoS attack that exploits the TCP three-way handshake by leaving connections incomplete is called:

a. SYN Flooding
b. ICMP Flood
c. Ping of Death
d. UDP Flood

Which tool is specifically known for performing DoS attacks by sending partial HTTP requests

and never completing them?

a. Hydra
b. Wireshark
c. Slowloris
d. Crunch

Correct Answer: c

Which mode allows a network interface card (NIC) to capture frames that are not specifically addressed to the host?

a. Monitor mode
b. Transparent mode
c. Promiscuous mode
d. Bridge mode

Correct Answer: c

Which of the following is/are examples of computer-based social engineering attacks?

a. Phishing emails
b. Pop-up window scams
c. Shoulder surfing
d. Chain letter emails
e. Tailgating

Correct Answer: a, b, d

What is the primary purpose of ARP spoofing in the context of sniffing attacks?

a. To increase the bandwidth available to the attacker.
b. To redirect traffic between two hosts through the attacker's machine for interception.
c. To encrypt network traffic for secure transmission.
d. To permanently disable a target host's network interface.

Correct Answer: b

Assignment- Week 10

Which of the following are examples of hardware-based attacks?

a. Physical probing
b. SQL injection
c. Cross-site scripting
d. Electromagnetic emission analysis
e. Phishing

Correct Answer: a, d

Which of the following statement(s) is/are true regarding side-channel attacks?

a. They exploit weaknesses in the implementation of an algorithm, not the algorithm itself.
b. They may require physical access to (or close proximity to) the target device, depending on the side channel being exploited.
c. They rely solely on analyzing input/output pairs of the algorithm.
d. They can exploit information such as timing, power consumption, or electromagnetic emissions.
e. They exploit mathematical weaknesses in the cryptographic algorithm's design.

Correct Answer: a, b, d

For modular exponentiation of x22 using the square-and-multiply method, how many squaring and multiplication operations are required, respectively?

a. 4 and 2
b. 3 and 2
c. 4 and 3
d. 5 and 2
e. 3 and 3

Correct Answer: a

Which of the following statement(s) is/are true regarding Differential Power Analysis (DPA) and Simple Power Analysis (SPA)?

a. DPA requires only a single power trace measurement.
b. DPA requires multiple power trace measurements combined with statistical analysis.
c. DPA uses statistical analysis of multiple traces to reveal information that may not be apparent from individual traces.
d. DPA is generally less effective than SPA.
e. SPA relies on directly observing patterns in a single or few traces.

Correct Answer: b, c, e

Which property of a PUF refers to the requirement that, given a PUF and an input (challenge) x, its corresponding output (response) y = PUF(x) can be computed efficiently?

a. Unclonability
b. One-wayness
c. Evaluability
d. Unpredictability
e. Tamper-evidence

Correct Answer: c

_____ is the full form of the acronym PUF used in hardware security.

a. Physically Unique Function
b. Physically Unclonable Function
c. Protected Unclonable Feature
d. Physically Untraceable Function

Correct Answer: b

What is a hardware Trojan?

a. A malicious software virus that specifically targets firmware update files.
b. A malicious, intentional modification to a chip's circuitry that activates under specific trigger conditions.
c. A testing tool used to validate hardware functionality before shipping.
d. A feature designed to make hardware run faster under heavy load.

Correct Answer: b

Which of the following statement(s) is/are true regarding hardware Trojans?

a. They typically incur small hardware overhead, making them difficult to detect.
b. They are generally easy to detect using a single, universal detection method.
c. Detecting hardware Trojans often introduces additional design, testing, or runtime overhead.
d. Hardware Trojans are exclusively inserted through software-level modifications.

Correct Answer: a, c

Which of the following hardware attack(s) are generally considered invasive in nature?

a. Black-box testing
b. Physical probing
c. Side-channel power analysis
d. Timing analysis

Correct Answer: b

Which of the following software-level technique(s) can help mitigate timing-based side-channel attacks?

a. Introducing redundant/dummy computations to equalize execution time across branches.
b. Masking sensitive intermediate data values during computation.
c. Deliberately increasing overall program complexity without addressing timing variance.
d. Ensuring constant-time implementations for security-critical operations.

Correct Answer: a, d

Assignment- Week 11

Which of the following Metasploit modules is used to take advantage of a known vulnerability on a target system?

a. Payload
b. Exploit
c. Encoder
d. Auxiliary

Correct Answer: b

Which command is used to launch the Metasploit Framework console?

a. msfvenom
b. msfconsole
c. metasploit-cli
d. msf-start

Correct Answer: b

Which Metasploit command is used to check which target operating systems are compatible with a selected exploit?

a. show targets
b. set targets
c. check compatibility
d. show exploits
e. show payloads

Correct Answer: a

Which Metasploit option is used to specify the target's IP address (the remote host to be attacked)?

a. set LHOST
b. set LPORT
c. set RHOST
d. set RPORT

Correct Answer: c

Which command in Metasploit's Meterpreter shell is used to retrieve stored user account password hashes from the target system?

a. Getuid
b. Getsystem
c. Hashdump
d. Sysinfo
e. ps

Correct Answer: c

Which tool is specifically designed to perform brute-force discovery of hidden files and directories on a web server?

a. SQLMAP
b. Dirb
c. Hydra
d. Crunch
e. Hashcat

Correct Answer: b

If a web page is vulnerable to error-based SQL injection, what is the expected behavior upon incorrect/malicious user input?

a. It will display a database error message revealing information about the query structure.
b. It will silently return no output at all.
c. It will redirect the user to a login page automatically.
d. It will crash the entire web server permanently.

Correct Answer: a

Which SQLMAP option is used to list all database names available on the target system?

a. --tables
b. --dbs
c. --dump
d. --current-db
e. --passwords

Correct Answer: b

Which of the following is/are recognized types of Cross-Site Scripting (XSS) attacks?

a. Stored XSS
b. Reflected XSS
c. DOM-based XSS
d. SQL-based XSS
e. Cookie-Injection XSS

Correct Answer: a, b, c

Consider the following simplified query used by a login form:
SELECT * FROM users WHERE username = 'INPUT1' AND password = 'INPUT2'
Which of the following input(s) for the username field, if unsanitized, would represent a classic SQL injection attempt to bypass authentication?

a. admin
b. admin' OR '1'='1

c. admin123
d. 'OR 1=1 –
e. guest_user
Correct Answer: b, d

Assignment- Week 12

In an ICMP Echo sweep, if the scanner receives an ICMP Type 0 packet in response to an ICMP Type 8 request, what does this indicate?
a. The target host is alive/up
b. The target host is down
c. The target port is filtered
d. The target host does not support ICMP
Correct Answer: a
Which of the following NMAP option(s) can be used to perform a TCP-based sweep for host discovery?
a. -PE
b. -PS
c. -PA
d. -PU
e. -PM
Correct Answer: b, c
Which of the following are automatically performed when using the NMAP -sn option?
a. ICMP Echo sweep
b. TCP sweep
c. UDP sweep
d. All of these
Correct Answer: a, b
Which NMAP option treats all hosts as online, effectively skipping the host discovery phase?
a. -sL
b. -sP
c. -PO
d. -sU
e. -Pn
Correct Answer: e
Consider the following statements:
(i) -sn performs host discovery only, without any port scanning.
(ii) -Pn skips host discovery entirely and proceeds directly to port scanning.
a. Only (i) is true
b. Only (ii) is true
c. Both (i) and (ii) are true
d. Both (i) and (ii) are false
Correct Answer: c
Which of the following statement(s) is/are true regarding NMAP's -F (fast scan) option?
a. It scans the top 100 most common ports.
b. It scans all 65535 ports.

c. It is faster than a default scan because it checks fewer ports.
d. It is equivalent to using --top-ports 1000
Correct Answer: a, c

Which NMAP option is specifically used to conduct a UDP port scan?

a. -sT
b. -sS
c. -sU
d. -sA
e. -sN

Correct Answer: c

Which scanning technique completes the full TCP three-way handshake with the target port to determine its state?

a. TCP SYN scan
b. TCP Connect scan
c. FIN scan
d. NULL scan
e. Xmas scan

Correct Answer: b

How many ports will be scanned using the NMAP command "nmap --top-ports 20 target_ip"?

Correct Answer: 20

How many total IP addresses will be scanned when executing the command nmap -sL 10.10.5.100-115?

Correct Answer: 16